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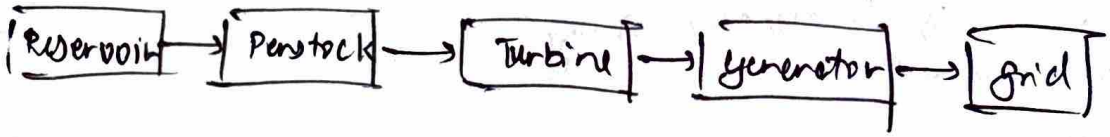
EXCS 'A'

REM
Assignment 4

Q1]

Principle of Hydro power Generation

- Water stored at height possesses potential energy
- Flowing water rotates turbine
- Turbine drives generator
- Electric power is transmitted to grid



Types of Hydropower plants

- Run of river plants
- Storage (reservoir) plants
- Pumped storage plants

Q2]

Biomass Conversion Technologies

- Combustion
- gasification
- Anaerobic digestion
- Pyrolysis
- Biofuel production

Q3]

Steps of Energy Management Systems

- Energy Policy
- Energy Audit
- Data Analysis
- Implementation
- Monitoring

Q4]

Economic Aspects of Renewable Energy

- Capital investment
- Operating cost
- Return on investment
- Financial viability

Q5]

Role in sustainable Development

- environmental protection
- Economic Growth
- Social development through rural electrification
- Reduced fossil fuel dependancy

Techno-Managerial Decision Support

- Helps evaluate technology choices
- Supports cost-benefit analysis
- Assists long-term infrastructure planning
- Guides sustainable investment